

for applying functions to the text such as NER and POS tagging.

```
#reading and tokenizing text
some_text = "This is some example text
that I will be using to demonstrate the
features of spacy."
read_text = myspacy(some_text)
print([token.text for token in read_
text])
```

The following will be the output of the code:

```
['This', 'is', 'some', 'example',
'text', 'that', 'I', 'will', 'be',
'using', 'to', 'demonstrate', 'the',
'features', 'of', 'spacy', '.']
```

We can also read text from a file as follows:

```
#reading and tokenizing text from a file
file_name = 'sample_text.txt'
sample_file = open(file_name).read()
read_text = myspacy(sample_file)
print([token.text for token in read_
text])
```

Sentence detection

A key feature of NLP libraries is detecting sentences. By finding the beginning and end of a sentence, you can break down text into linguistically meaningful units, which can be very important for applying machine learning models. It also helps you in applying parts of speech tagging and named entity recognition. spaCy has a *sents* property that can be used for sentence extraction.

```
#sentence detection
sample_passage = "This is an example
of a passage. A passage contains many
sentences. Sentences are denoted using
the dot sign. It is important to detect
sentences in nlp."
read_text = myspacy(sample_passage)
sentences = list(read_text.sents)
for sentence in sentences:
    print(sentence)
```

As you can see in the following output, we have successfully broken down the *sample_passage* into discernible sentences.

```
This is an example of a passage.
A passage contains many sentences.
Sentences are denoted using the dot
sign.
It is important to detect sentences in
nlp.
```

Removing stop words

An important function of NLP is to remove stop words from the text. Stop words are the most commonly repeated words in a language. In English, words such as 'are', 'they', 'and', 'is', 'the', etc, are some of the common stop words. You cannot form sentences that make semantic sense without the usage of stop words. However, when it comes to machine learning, it is important to remove stop words as they tend to distort the word frequency count, thus affecting the accuracy of the model.

spaCy has a list of stop words in its library for English. To be precise, there are 326 stop words in English. You can remove them from the text using the *is_stop* property of spaCy.

```
#removing stopwords
print([token.text for token in read_
text if not token.is_stop])
```

After removing the stop words, the following will be the output of our sample text.

```
['example', 'passage', '.',
'passage', 'contains', 'sentences',
'.', 'Sentences', 'denoted', 'dot',
'sign', '.', 'important', 'detect',
'sentences', 'nlp', '.']
```

Lemmatisation of text

Lemmatisation is the process of reducing the inflected forms of a word such that we are left with the root of the word. For example, 'characterisation', 'characteristic' and 'characterise',

are all inflected forms of the word 'character'. Here, 'character' is the lemma or the root word. Lemmatisation is essential for normalising text. We use the *lemma_* property in spaCy to lemmatise text.

```
#lemmatisation of text
for word in read_text:
    print(word, word.lemma_)
```

The following is the lemmatised output of the sample text. We output the word along with its lemmatised form. To preserve page space, I am sharing the output of a single sentence from our sample text.

```
Sentences sentence
are be
denoted denote
using use
the
dot dot
sign sign
. .
```

Finding word frequency

The frequency at which each word occurs in a text can be vital information when applying a machine learning model. It helps us to find the main topic of discussion in a piece of text and helps search engines provide users with relevant information. To find the frequency of words in our sample text, we will import the *Counter* method from the *collections* module.

```
#finding word frequency
from collections import Counter
word_frequency = Counter(read_text)
print(word_frequency)
```

The following is the output for the frequency of words in our sample text:

```
Counter({'This': 1, 'is': 1, 'an': 1,
example: 1, 'of': 1, 'a': 1, 'passage': 1,
.: 1, 'A': 1, 'passage': 1, 'contains': 1,
many: 1, 'sentences': 1, '.': 1, 'Sentences':
1, 'are': 1, 'denoted': 1, 'using': 1, 'the':
```